Recall

An open <u>ball</u> in $\mathbb{R}^n$ with center $\underline{x}^0$ and radius δ is the set of points

$$B(\underline{x}^0, \delta) = \{\underline{x} \in \mathbb{R}^n : d(\underline{x}, \underline{x}^0) < \delta\}.$$

Def'ns

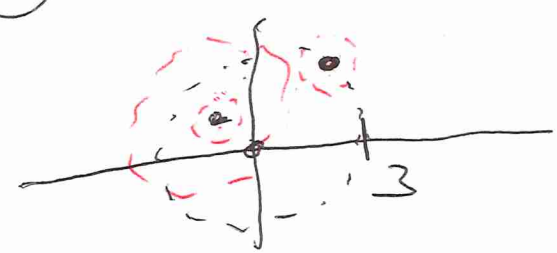
An <u>open set</u>, $\mathcal{O}$, is a set for which if $\underline{x} \in \mathcal{O}$, we can find $B(\underline{x}, \delta)$ such that $B(\underline{x}, \delta) \subset \mathcal{O}$.

A <u>closed set</u>, A, is a set whose complement, A^c, is open

A <u>boundary pt</u>, $\underline{x}$, of A is a pt ~~so the~~ for which <u>all</u> balls centered at $\underline{x}$, $B(\underline{x}, \delta)$ contains points in A and A complement, A^c.

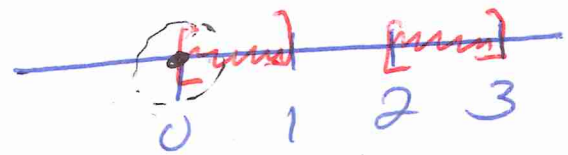
∂A = set of all boundary pts.

ex) $B((0,0), 3) \subset \mathbb{R}^2$



The <u>closure of a set</u> is the set plus all of its boundary pts. $\bar{A} = A \cup \partial A$

ex) $A: 0 \le x \le 1$ or $2 \le x \le 3$



Is A open or closed? or neither?

What are the points in ∂A? $\{0, 1, 2, 3\}$

What is $\bar{A}$ (closure)? $\{x \in \mathbb{R} : 0 \le x \le 1 \text{ or } 2 \le x \le 3\}$

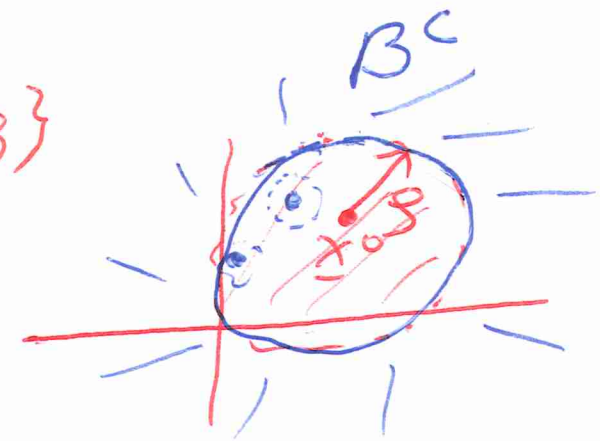
$\bar{A} = A$

for any closed set, $A = \bar{A}$.

ex) $\mathbb{R}^2$ set is $B(\underline{x}^0, \rho) = \{\underline{x} \in \mathbb{R}^2 : d(\underline{x}, \underline{x}^0) < \rho\}$

Find open or closed? or neither?

$B(\underline{x}^0, \rho)^c = \{\underline{x} \in \mathbb{R}^2 : d(\underline{x}, \underline{x}^0) \geq \rho\}$



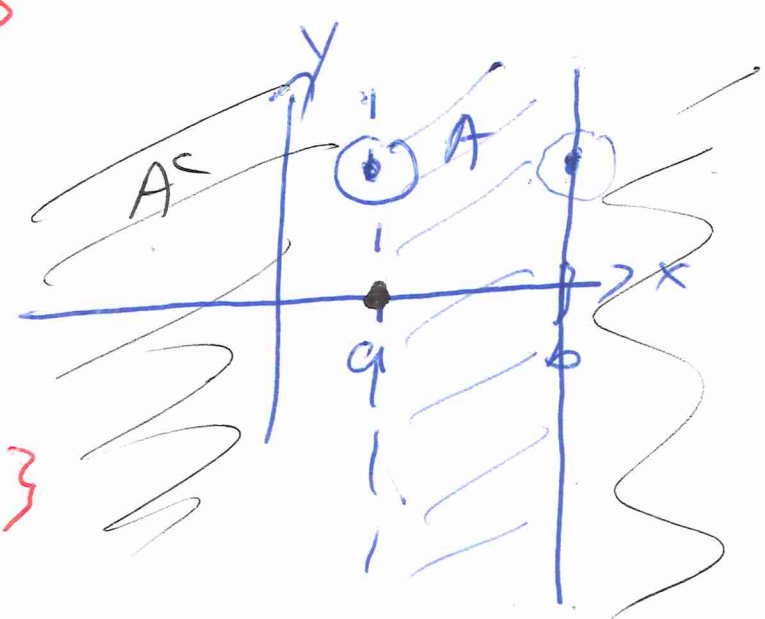
$B \neq \overline{B(\underline{x}^0, \rho)} = \{\underline{x} \in \mathbb{R}^2 : d(\underline{x}, \underline{x}^0) \leq \rho\}$

such that

$\partial B(\underline{x}^0, \rho) = \{\underline{x} \in \mathbb{R}^2 : d(\underline{x}, \underline{x}^0) = \rho$

ex) $\mathbb{R}^2$ $A = \{(x,y) \in \mathbb{R}^2 : a < x \leq b\}$

Is A open, closed, neither?



Find

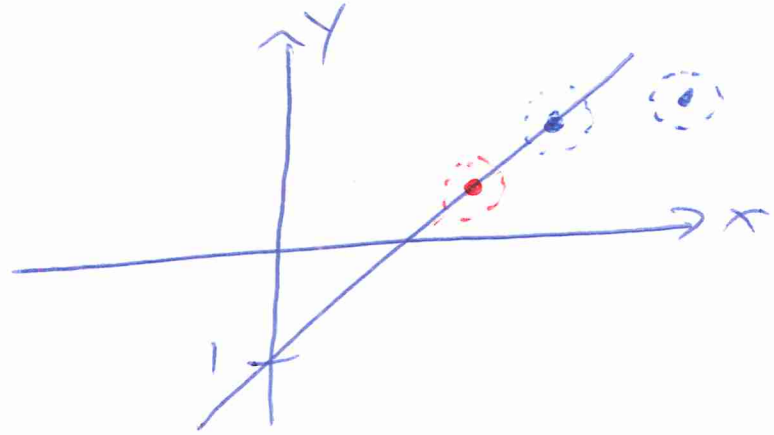
$\partial A = \{(x,y) \in \mathbb{R}^2 : x = a \text{ or } x = b\}$

$A^c = \{(x,y) \in \mathbb{R}^2 : x \leq a \text{ or } x > b\}$

$\bar{A} = \{(x,y) \in \mathbb{R}^2 : a \leq x \leq b\}$

ex) $\mathbb{R}^2$, A = line $y = x - 1$

open, closed, neither?



$A^c = \{(x,y) \in \mathbb{R}^2 : y \neq x - 1\}$

$\partial A = \{(x,y) \in \mathbb{R}^2 : y = x - 1\}$ $(x=0, y=1)$

$\bar{A} = \{ \quad '' \quad \}$

$A = \{(x,y) \in \mathbb{R}^2 : y = x - 1\}$

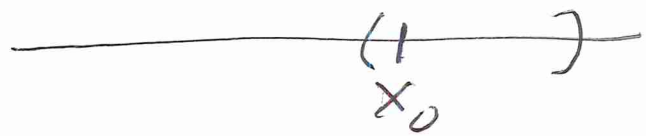
$A = \mathbb{R}$

Def'n A <u>neighborhood</u> of $\underline{x}$, is any open set containing $\underline{x}$.

ex) sol'n to a nonlinear first-order ODE

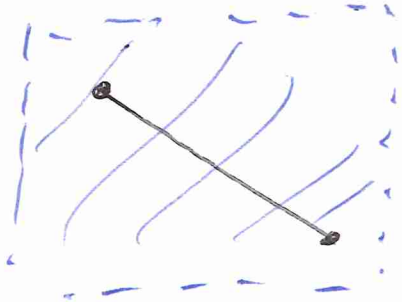
$$y' = f(x,y) \qquad y(x_0) = y_0 \quad \ldots \text{other conditions} \ldots$$

is only guaranteed to exist in a neighborhood (nbd) of x_0.

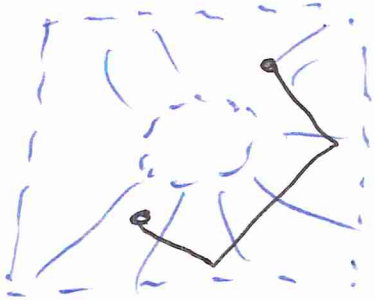


Def'n An open set in $\mathbb{R}^n$ is <u>connected</u> if any 2 pts in ~~The~~ the set can be connected by a polygonal path which is contained in the set.

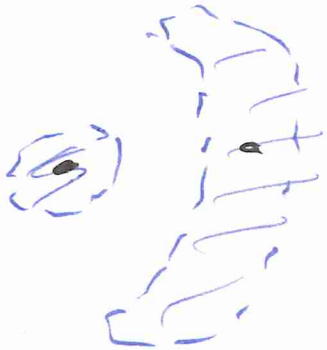
ex)



connected



connected



Not connected.

~~ex~~

<u>Def'n</u> Ω is a <u>domain</u> if it is open and connected.

Notation

$$\frac{\partial f}{\partial x} = f_x = D_1 f$$

1st indep. variable

$$\frac{\partial f}{\partial y} = f_y = D_2 f$$

$$\frac{\partial^2 f}{\partial x^2} = f_{xx} = D_1^2 f$$

$$\frac{\partial^2 f}{\partial y^2} = f_{yy} = D_2^2 f$$

$$\frac{\partial^2 f}{\partial x \partial y} = f_{xy} = D_{12} f$$

If $f(x,y)$ is smooth enough mixed partials commute

$$f_{xy} = f_{yx}$$

$$y = f(x)$$

dep. indep.

$$\frac{dx}{dt}$$

dep. indep.

$$\boxed{\frac{\partial z}{\partial x} = e^y}$$

$$\frac{\partial z}{\partial x} = 5$$

dep. variable z

indep. variable x, y